

Michael Boiman

Freelance Quality Engineer & AI Architect

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Career Profile

Quality Engineer and AI Architect. Quality engineering and AI architecture are not consecutive chapters here, they are two tracks running in parallel and feeding into each other.

In enterprise projects (**DB Vertrieb**, **DVAG**, **TÜV Süd**), quality monitoring systems took shape, AI-driven test automation was added, and legacy migrations were secured with 24/7 validation.

Today that means e-invoicing to EN 16931 in production and a network of agents consulting each other over the open MCP and A2A protocols.

I also pass this knowledge on, in developer and business workshops and in an invited talk at TU Darmstadt, translating hands-on AI for technical and non-technical audiences alike. I have introduced doctors, lawyers and financial advisers to working with AI in their own daily work and supported business teams across several projects.

EN 16931
e-invoicing in production
Peppol BIS 3.0 into SAP ByDesign

MIT
open-source agent infrastructure
github.com/bks-lab/open-bridge

ISTQB®
Certified Tester
test method to the international standard

Competencies

Quality Engineering	Quality Engineering, Specification & Verification Design, Playwright, Behave / pytest (BDD), Cucumber/Gauge, JMeter, Gatling
AI & Agents	Claude Code (Skills, Hooks, MCP), Agent Harness Engineering, LLMs (Claude, Gemini, GPT), Local LLMs (vLLM, Ollama), LLM Evaluation & Benchmarks, Hugging Face (Spaces, Datasets), Agent Orchestration (MCP & A2A), a2a-sdk & Agent Cards, Google ADK, LangChain
Delivery & Automation	Python, TypeScript, FastAPI, JSON-RPC 2.0 & SSE, CI/CD, Docker & Container, REST APIs, Agile (Scrum/Kanban)
Platform & Monitoring	Peppol/E-Invoicing (EN 16931), Azure Functions, Azure DevOps, Azure Monitor & KQL, Cloudflare (Workers, Pages, R2), Microsoft 365 / Exchange Online, Grafana, Elasticsearch, Kibana, OpenTelemetry / OTLP
More	AWS, Google Cloud, VMware, PowerShell, LoadRunner (VuGen, Web HTTP/HTML, TruClient, Citrix), SoapUI, Jira with Zephyr / Xray, Dynatrace, Jenkins, Instana

Professional Experience

AI Test Automation Engineer · Building the AI Test Automation Process

09/2026-10/2026

KION Group

AI-driven test automation for one of the group's customer portals: building the complete process, from deriving the test cases to a green CI pipeline.

Key Responsibilities

- **Test orchestrator**: a chain of AI building blocks with a fixed log schema, its own model tier per block and usage tracking, packaged as a reusable command for the team
- **Architecture rules for the test code base**: derived from a multi-stage analysis with several AI agents, independently cross-checked and documented
- **Architecture check for generated test code**: detects violations of the team's style guide and fixes the minor ones on its own
- **Calibrated comparison runs**: AI-generated E2E tests compared on cost, usage and conformance to the standard, across several model tiers

Tools & Technologies

- Testing: Playwright, TypeScript, Angular code base
- AI: Claude Code (several model tiers), GitHub Copilot CLI
- CI/CD & tooling: GitHub Actions, Sonar, Jira with Xray, Confluence, SAFe
- Scripting & containers: PowerShell, Docker

Senior Quality Engineer · Test Automation and Monitoring

09/2025-09/2026

TÜV Süd

Quality Engineering for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

Key Responsibilities

- **BDD API Test Framework**: Development of comprehensive test framework with Python/Behave, self-healing authentication, HTML report generator with embedded API responses
- **E2E Test Automation**: Playwright E2E test suite for complete user journey (Setup -> Upload -> Transformation -> Download) with 500 error handling and route mocking
- **Performance & Monitoring**: Development of Azure Monitor Workbooks (QA Live Testing Companion, Error Investigation Assistant) with KQL for end-to-end monitoring
- **Bug Analysis & Quality Intelligence**: Implementation of comprehensive bug tracking system with WIQL queries, automated dashboards, and executive PDF reporting

Tools & Technologies

- Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures
- Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics
- Backend: FastAPI, Python, Pydantic, SQLModel, Azure Functions, Azure Service Bus
- Frontend: React, TypeScript, Vite, TanStack Router
- AI Integration: Azure OpenAI (GPT-4o), LangChain, Azure AI Search, Embeddings
- DevOps: Azure DevOps Pipelines, Git, Docker, PowerShell, Azure Blob Storage, Poetry
- Agentic Coding: GitHub Copilot CLI

LLM Infrastructure Architect & Automation Engineer

since 06/2025

BKS · AI Research & Development (internal)

Development of LLM-orchestrated Enterprise Development Ecosystem: Fully integrated system for knowledge management, project automation, and development workflows with Claude, Gemini, and OpenAI GPT.

Key Responsibilities

- **Open-Source Agent Infrastructure**: open-bridge under the MIT licence, public at github.com/bks-lab/open-bridge, with a live demo site. The generic CORE layer (agent runtime, MCP gateway, skill tree, promote flow with a content-leak gate in CI) grows out of my own instance and is published upward by scope, with its own test suite and continuous releases since June 2026.
- **Test bench for open models (lab-bench)**: own test bench on which BKS-Lab measures which model can carry a task on hardware it controls. Open models (including Qwen3, DeepSeek-R1, Nemotron) run self-hosted through vLLM and Ollama on an RTX 4090, alongside GLiNER encoders, a classically trained classifier and a hosted decision API for comparison. Measured among others: German retrieval for RAG, speech to text, invoice fields from page images, routing and slot extraction, with 95 % bootstrap intervals, McNemar tests, a sanity check against published reference values, speed, GPU memory and energy. Nightly runs in a Proxmox lab with Prometheus and Grafana; results are public as a dataset and Space on Hugging Face (bks-lab/lab-bench-results, bks-lab/lab-bench)
- **Harness engineering**: what an agent achieves is decided mostly outside the model: context, tool selection, permissions, state across sessions, verification. That layer is measured here rather than estimated. What a session loads before it answers the first word carries a declared ceiling that a CI step checks in bytes; rules that rarely bite load on their trigger word instead of at start-up; the project registries sit in context as a card and are fetched one entry at a time; a further check reports what exists in the tree but is no longer reachable from any file a session loads, and attacks itself to prove it still catches things. The tools behind this are public and run in public CI
- **Data separation as architecture, not as agreement**: separate instances per client, a rule forbidding access between instances, and a fail-closed contract for materialising shared configuration (the sync aborts rather than writing on doubt). Protection against data leakage at three points: pre-commit hook, pre-push hook and three validators in CI. Plus an adversarial simulation that attacks the protection itself on a schedule, as its own CI job in a sealed sandbox

Tools & Technologies

- LLM Orchestration: Claude (primary), Google Gemini, OpenAI GPT, Model Context Protocol (MCP), Google Agent Development Kit (ADK)
- AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration; Cursor before that
- Multi-Agent: A2A Protocol 1.0, a2a-sdk, Agent Cards, MCP-to-A2A gateway, JSON-RPC 2.0, SSE Streaming
- Backend: Python 3.13, FastAPI, uvx Distribution, Git Submodules
- Automation: GitHub Actions, GitHub GraphQL API, gh CLI
- Integration: SharePoint API, Microsoft Graph, Exchange Online, Elasticsearch, Kibana, OpenTelemetry/OTLP
- Operations: Cloudflare (Workers, Pages, R2, DNS), Tailscale, launchd, restic/rclone
- Cloud: AWS, Google Cloud (alongside Azure)
- Development: Bash Scripting, jq, Docker, Multi-Agent Systems; orchestrated repos span Python, TypeScript, Rust and Go

AI-driven QA Environment for Energy Infrastructure

07/2024-01/2025

AkkuSwap Startup

QA Leadership for EU-wide Battery Swap Infrastructure: AI-driven simulation tool for battery swap station network with energy infrastructure integration

Technical Stack

- Infrastructure: Linux, Docker, Azure OpenAI
- Automation: pytest, AI code generation integration patterns
- Monitoring: Grafana, Azure Monitoring
- Energy Systems: Battery swap infrastructure, grid integration simulation

Key Achievements

- Established QA methodology for AI-driven infrastructure simulation
- Created testing framework for energy sector critical systems
- Validated proof-of-concept with measurable reliability improvements

Technical Lead & AI Automation Architect

01/2024-04/2025

BKS on behalf of a client in the digital economy

Technical lead & development: Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform (under NDA). Full responsibility from requirements analysis through implementation to production go-live.

Focus areas

- **Workflow automation**: Extraction, analysis and classification of incoming invoices using Python and Azure Functions.
- **E-invoice classification & data extraction**: Full processing in accordance with EU standard EN 16931 (ZUGFeRD, XRechnung, Factur-X), including structural and content parsing.
- **Peppol Network Integration**: Connection to European e-invoicing network through a certified Peppol Access Point for automated invoice receipt and dispatch with UBL/CII formats.
- **SAP ByDesign Integration**: Automatic transfer of validated invoice data to SAP ByDesign for seamless ERP integration.

Tools & technologies

- Azure Functions: Serverless orchestration of automation workflows.
- Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance.
- E-Invoicing: Peppol Network, Access-Point-API, ZUGFeRD/Factur-X, XRechnung, UBL 2.1, CII
- ERP Integration: SAP ByDesign, Salesforce
- Microsoft Graph API: Access to Outlook mailboxes and SharePoint libraries.
- GitHub: Version control, CI/CD pipelines and collaborative development.
- Kibana: Monitoring, log analysis and performance dashboards.
- Streamlit: For developing an interactive web application for visualization.
- AI Automation: Claude AI, Custom Skills Development, Git Analysis, Natural Language Processing
- Protocols: Model Context Protocol (MCP), Agent-to-Agent (A2A), JSON-RPC 2.0

AI Engineering Lead & ML Solutions Architect

04/2023-04/2024

BKS on behalf of a publishing house

Project management & technical lead: Development and deployment of an AI-driven email bot to automate customer communication, AI-assisted processing of incoming emails and triggering of follow-up activities in existing SAP systems. Responsible for end-to-end project coordination and team management from idea to implementation.

Tools & technologies

- Azure Functions: For serverless application architectures and ML model execution.
- Python: Main programming language for the email bot and ML models.
- Microsoft Graph API: For accessing and processing emails in Microsoft Outlook 365.
- SAP API: For querying customer data and communicating with SAP systems.
- GitHub: For version control, CI/CD pipelines and collaborative development.
- Kibana: For real-time monitoring and system performance analysis.
- Streamlit: For developing an interactive web application for system monitoring.

Platform Quality Architect (CI/CD, Monitoring, Automation)

08/2021-05/2025

DVAG

Test management & coordination, as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams.

Result: Real-time sprint-quality transparency via Grafana dashboards, quality-gated releases in the teams' CI/CD pipelines: data-driven release decisions replacing manual data gathering.

Tools

- IntelliJ / Visual Studio Code
- Java / Kotlin / JavaScript / TypeScript
- Playwright / Gauge / Karate
- Python for processing and analysis of test and reporting data
- SQL for analysing and validating test data
- CI/CD / GitHub (reusable workflows)
- Helm / Docker / Containers
- PowerShell
- JIRA / Confluence / Zephyr
- Grafana
- JMeter / Gatling / LoadRunner (VuGen, Web HTTP/HTML)
- Behaviour Driven Development
- Consumer-Driven Contracts (Pact.io)

Founder: Boiman.Solutions and QualityCluster

06/2021-08/2021

Boiman.Solutions

Set up his own consultancy Boiman.Solutions and the quality engineering community QualityCluster, a group of quality engineers working towards more quality through a higher degree of automation.

DevOps Quality Lead & Dashboard Architect

01/2017-05/2021

DB Vertrieb

Test management, test coordination, test automation, agile (Kanban, Scrum, SAFe): Coordination as PO of a cross-team QA team including cross-team onboarding, support, consulting, training

Tools

- IntelliJ
- Java
- Cucumber / Cypress
- JMeter / Gatling
- CI/CD / GitLab CI / Jenkins
- Helm / Docker
- JIRA / Confluence
- Kibana / Grafana / Instana / Graylog
- AWS
- Behaviour Driven Development
- Consumer Driven Contracts (Spring Cloud Contract)
- Ionic (hybrid app)
- Selenium

Earlier positions

09/2015-01/2017	**Performance Engineering Lead & Test Architect** DB Systel
04/2015-09/2015	**Senior iOS Developer & Mobile Architect** Telekom
05/2009-03/2015	**Quality Assurance Lead & Test Automation Architect** Siemens
03/2009-05/2009	**Performance Test Engineer & Load Testing Specialist** ING-DIBA
09/2005-12/2008	**Performance Test Engineer & Quality Consultant** SQS AG · C1 SetCon (Performance Engineering) · freelance

Selected projects

E-Invoicing Automation Platform

Python · Azure Functions · Peppol · Peppol Access Point · SAP ByDesign

Production e-invoicing pipeline compliant with EU standard EN 16931.

- **Problem:** Inbound and outbound invoices in multiple formats (ZUGFeRD, XRechnung, UBL) required EN 16931-compliant processing.
- **Solution:** Peppol connection through a certified access point, hexagonal architecture, automatic transfer into the ERP.
- **Result:** Production system, invoice processing runs automated end-to-end into the ERP. Seven repositories, roughly 2,900 automated test cases, more test code than production code, an enforced coverage threshold per repository.

24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering

Java · Cucumber · Kibana · Graylog · Elasticsearch

Dual-run validation for safely migrating a mainframe price calculation.

- **Problem:** Millions of price combinations had to migrate without quality loss or data errors.
- **Solution:** 24/7 validation with randomly generated tests, live dashboard and full deviation traceability.
- **Result:** Safe migration without data loss, all combinations validated around the clock.

Agent mesh and MCP-to-A2A gateway

Python · a2a-sdk · A2A Protocol 1.0 · Agent Cards · JSON-RPC 2.0

Three agents, one translator, one open protocol. The agent on my CV page is one of them.

- **Problem:** An agent that only holds its own knowledge guesses at everything outside it.
- **Solution:** Three agents, each with its own subject, consult each other over the open A2A protocol, read-only. In front of them a stateless gateway offering the same agents to MCP clients as three tools.
- **Result:** Publicly retrievable agent card, 98 automated test cases, generic framework open under MIT.

Quality Dashboard · Real-time Overview

Grafana · Elasticsearch · Python · CI/CD · Playwright

Real-time quality overview for enterprise go/no-go decisions.

- **Problem:** Release decisions required hours of gathering data from the scattered test tools of several teams and environments.
- **Solution:** End-to-end pipeline with live dashboard, multi-source integration and automated PDF reports.
- **Result:** Data-driven release decisions in real time instead of manual collection.

AI-driven Test Automation

Python · OpenAI GPT-4 · Playwright · IDE Integration · Jira API

Guideline-based AI agent in the IDE that generates tests from Jira stories.

- **Problem:** Deriving test cases from a story is manual work, and a model without constraints invents steps and selectors the application does not have.
- **Solution:** The agent reads the story over the Jira API; the project's own conventions (page objects, naming rules, permitted selectors) sit beside it as a rule file and bound what it may produce.
- **Result:** The test comes out in the style of the existing Playwright project. Typing an empty file turns into reviewing a draft.

Appointment enquiry over WhatsApp and phone

Python · FastAPI · WhatsApp Business Cloud API · Speech-to-Text · Text-to-Speech

Two channels on one domain core, WhatsApp and phone, reading free slots live from the booking portal.

- **Problem:** Calls were lost during work, and a language model must never invent an appointment.
- **Solution:** The model runs the conversation, the free slots come live from the portal API. It pencils an appointment in but never books it.
- **Result:** The WhatsApp channel has been in production at the client since 08/2026 with real end users and data residency in Germany; consent comes before the first substantive sentence.

AI Context Orchestrator: Multi-Repo Development Platform

Python · YAML · Claude Code Plugins · Hooks API · GitHub Actions

Central control system for AI-assisted development across customer, partner and internal repos.

- **Problem:** Development across many repos and multiple customers required constant context switching and manual coordination.
- **Solution:** Orchestration hub with a three-way split registry, one skill tree for every tool and a hook-based permission system.
- **Result:** Workflows from incident response to meeting documentation largely automated.

Talks & workshops

04/2026	**Guest Lecture: AI-Assisted Software Development in Practice** TU Darmstadt · Information Systems Group · Lab course “AI Startup: From Idea to Execution”
06/2025	**Workshop: Development with Generative Language Models** Developer Workshop · Agent-based Software Development
04/2025	**Presentation: Efficient Documentation through Automation** Enterprise Presentation · AI-powered Documentation Workflows
2023	**Applications of AI in Business Context** AI Workshop for Business Integration

Education

2000-2005	**Diploma in Computer Science (UAS)** Cologne University of Applied Sciences, Gummersbach
2005	**ISTQB Certified Tester**
2007	**Certified Dynatrace Diagnostics Basic Training**

Languages

German	Native
English	Fluent (C1)